

Manifold-Informed Architecture Benchmark

BREAST_CANCER: 2 classes, 30D input

Generated: 2026-04-15 09:41:11 | Apple M5 Max MacBook Pro, 64 GB RAM, 2TB SSD

waverider @ 054030a (-abbrev-re 054030a600978c0e9ffac58faf7157939927d009) | 2026-04-14 22:20:05
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Best: Standard (240→120) test acc = 0.9731 ± 0.0141

Manifold: 30D → 9D (70.0% noise dimensions)

Experimental Setup

Parameter	Value
Dataset	BREAST_CANCER
Input dimensionality	30D
Classes	2
Intrinsic dim (d)	9
Variance threshold (τ)	0.9
Epochs	50
Trials	3
Batch size	32
Learning rate	0.001
k neighbors (local PCA)	40

Manifold Discovery

Local PCA over the training set, $k = 40$ neighbors. Intrinsic dimensionality d is taken as the maximum per-class maximum at $\tau = 0.9$, clamped to $n_{\text{classes}} = 2$.

τ	Mean d	Std	Min	Max	Noise (%)
0.95	10.4	0.5	9	12	65.2
0.90	8.2	0.6	7	10	72.5
0.85	6.9	0.6	5	8	77.1
0.80	5.9	0.5	4	7	80.2

Per-Class Intrinsic Dimensionality

Class	Mean d	Std	Min	Max
1	8.4	0.6	7	9
0	7.8	0.5	7	9

Architecture Comparison

Architecture	Params	Test Acc ($\mu \pm \sigma$)	Loss	Acc/Kparam
Euclidean KNN (k=7)	0	0.9596 ± 0.0142	—	—
ManifoldModel ($\tau=0.9$)	0	0.9631 ± 0.0161	—	—
	36,602	0.9731 ± 0.0141	0.1552	0.0266
Standard (240→120)★				
Manifold (2d→d, d=9)	299	0.9713 ± 0.0134	0.0808	3.2486
Wide Manifold (d+1=10)	332	0.9713 ± 0.0117	0.0864	2.9257
Manifold+ManifoldAdam (d=9)	299	0.9690 ± 0.0127	0.0882	3.2407
PCA→9D + MLP (2d→d)	371	0.9701 ± 0.0114	0.0839	2.6150
Intrinsic Dim (PCA→9D→C)	110	0.9696 ± 0.0115	0.1020	8.8141

★ = best performing architecture

Key Findings

- **Best architecture:** Standard (240→120) — test accuracy 0.9731 ± 0.0141
- **Manifold compression:** 30D → 9D (70.0% of ambient dimensions carry no structure)

Result Figure

Provenance

Item	Value
Repository	Flux-Frontiers/waverider
Commit	054030a (-abbrev-re 054030a600978c0e9ffac58faf7157939927d009)
Commit date	2026-04-14 22:20:05 -0400
Commit message	chore(release): bump version to 0.6.0
Python	3.12.13
TensorFlow	2.21.0
Device	CPU (forced)
Host	Turing
OS	macOS-26.4-arm64-arm-64bit
Machine	Apple M5 Max MacBook Pro, 64 GB RAM, 2TB SSD
Generated	2026-04-15 09:41:11

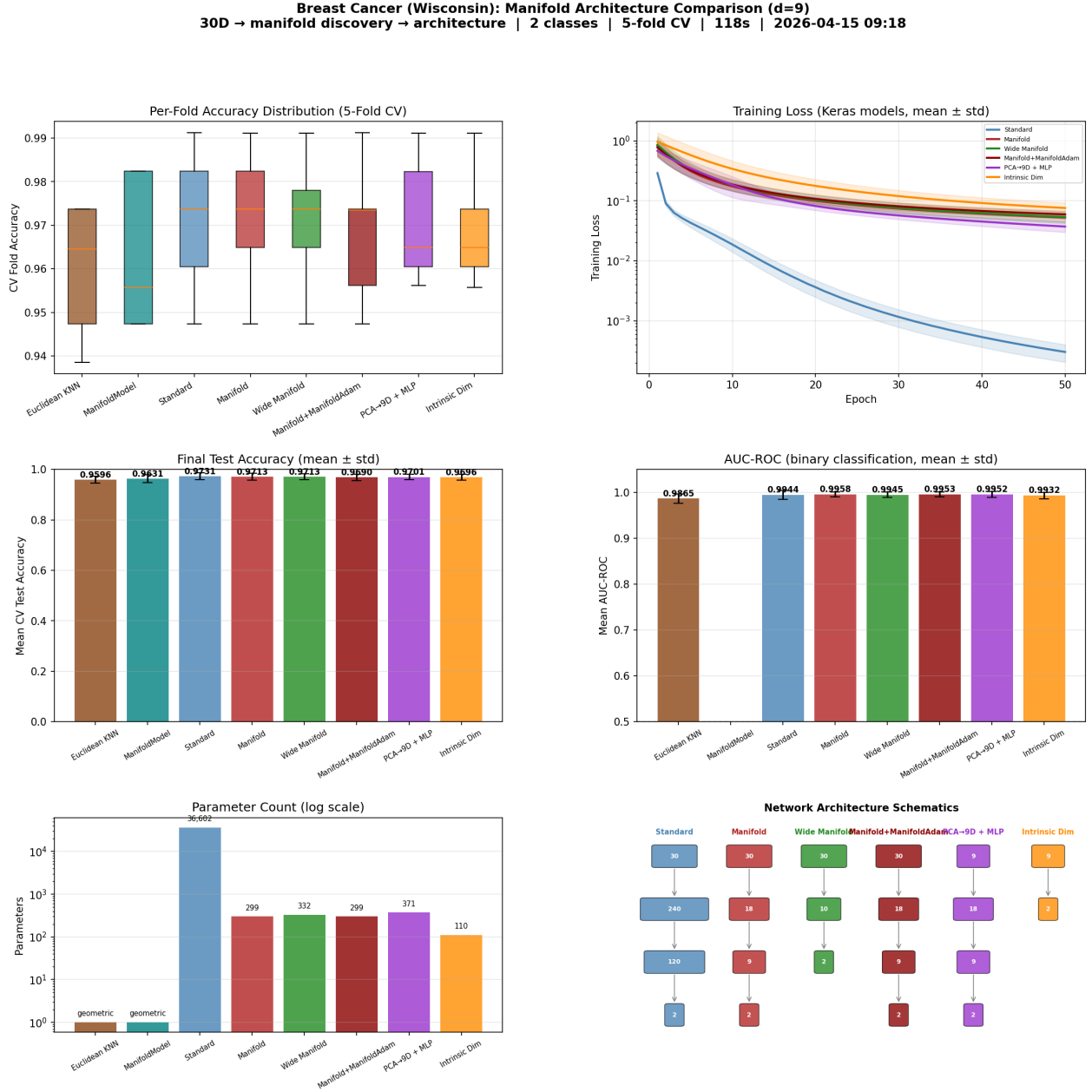


Figure 1: BREAST_CANCER manifold-informed vs. standard architecture comparison. $d = 9$, 50 epochs, 3 trials.